Distinct Distances in Planar Point Sets with No Four Concyclic Points

July 19, 2026

Abstract

For a set P of n distinct points in the Euclidean plane, let $d_P(p)$ denote the number of distinct distances from $p \in P$ to the other points of P. We study the least possible value of $\max_{p\in P} d_P(p)$ under the restriction that no four points of P lie on one circle. A direct multiplicity argument gives only $\lceil (n-1)/3 \rceil$. We show, using an elementary bound for the number of diameter pairs in a planar set, that the exceptional congruence class can also be improved, yielding

$$f(n) \geq \left\lceil \frac{n}{3} \right\rceil .$$

We then give an explicit construction on two perpendicular axes proving

$$f(n) \leq \left\lfloor \frac{3n}{4} \right\rfloor - \mathbf{1}_{\{4|n\}}.$$

Consequently $\limsup f(n)/n \leq 3/4$, so an asymptotic lower bound of the form $f(n) > (1 - o(1))n$ is impossible. The note is completely self-contained. We also formulate exactly what additional "distance-fibre deficiency" would be required to improve the lower bound to $(1/3 + c)n$.

Keywords. Distinct distances; concyclic points; diameter graph; extremal planar geometry.

1 Definition and main result

Throughout, a point configuration means a finite *set* of pairwise distinct points. Thus the condition that no four points are concyclic always refers to four distinct points and to an ordinary Euclidean circle.

Definition 1.1. Let $P \subset \mathbb{R}^2$ be finite and let $p \in P$. Define

$$d_P(p) := \left| \{ |p - q| : q \in P \setminus \{p\} \} \right| .$$

For $n \geq 1$, let

$$f(n) := \min_{\substack{P\subset\mathbb{R}^2,\ |P|=n \\ \text{no four points of } P \text{ are concyclic}}} \max_{p\in P} d_P(p).$$

The minimum is well defined: the class of admissible configurations is nonempty, and the quantity being minimized is an integer in $\{0, 1, \ldots, n-1\}$.

Theorem 1.2 (Main bounds). *For every integer $n \geq 2$,*

$$\boxed{\left\lceil \frac{n}{3} \right\rceil \leq f(n) \leq \left\lfloor \frac{3n}{4} \right\rfloor - \mathbf{1}_{\{4|n\}}.}$$

Equivalently, if $m \geq 0$, then the upper bound is

$$f(n) \leq \begin{cases} 3m - 1, & n = 4m \quad (m \geq 1), \\ 3m, & n = 4m + 1, \\ 3m + 1, & n = 4m + 2, \\ 3m + 2, & n = 4m + 3. \end{cases}$$

Here $\mathbf{1}_{\{4|n\}}$ denotes the indicator of the divisibility condition $4 \mid n$: it is 1 when 4 divides n and 0 otherwise.

Corollary 1.3. *One has*

$$\frac{1}{3} \leq \liminf_{n\to\infty} \frac{f(n)}{n} \leq \limsup_{n\to\infty} \frac{f(n)}{n} \leq \frac{3}{4}.$$

In particular, there is no sequence $\varepsilon_n \to 0$ such that

$$f(n) > (1 - \varepsilon_n)n$$

for all sufficiently large n. Thus the proposed estimate

$$f(n) > (1 - o(1))n$$

is false.

Proof. The limit inequalities follow immediately from theorem 1.2. The last assertion follows because the upper bound in theorem 1.2 gives $f(n)/n \leq 3/4$ for every $n \geq 2$, whereas $(1 - \varepsilon_n) > 3/4$ for all sufficiently large n whenever $\varepsilon_n \to 0$. ☐

The lower and upper bounds are proved separately in sections 4 and 5.

2 The elementary distance-fibre bound

Fix an admissible configuration P of cardinality n and a point $p \in P$. For each distance $r > 0$ realized from p, the corresponding distance fibre is

$$F_p(r) := \{q \in P \setminus \{p\} : |p - q| = r\}.$$

It lies on the circle of centre p and radius r. Hence the hypothesis immediately controls its cardinality.

Proposition 2.1 (Capacity bound)**.** *For every $p \in P$ and every realized distance $r > 0$,*

$$|F_p(r)| \leq 3.$$

Consequently

$$d_P(p) \geq \left\lceil \frac{n-1}{3} \right\rceil.$$

Proof. If $|F_p(r)| \geq 4$, then four distinct points of P lie on the circle with centre p and radius r, contrary to the hypothesis. The fibres $F_p(r)$ over the $d_P(p)$ realized distances partition the $n - 1$ points of $P \setminus \{p\}$. Therefore

$$n - 1 = \sum_r |F_p(r)| \leq 3d_P(p),$$

which gives the stated integer lower bound. ☐

For $n \not\equiv 1 \pmod 3$, this already equals $\lceil n/3 \rceil$. If $n = 3m + 1$, however, it gives only m rather than $m + 1$. The missing unit is forced by the geometry of diameter pairs.

3 A self-contained diameter-graph bound

Let $S \subset \mathbb{R}^2$ be a finite set with at least two points, and write

$$D(S) := \max\{|x - y| : x, y \in S\}.$$

The *diameter graph* $G_D(S)$ has vertex set S, with an edge xy precisely when $|x - y| = D(S)$.

We shall prove the planar bound

$$|E(G_D(S))| \leq |S|.$$

The proof is included in full because this is the only genuinely geometric ingredient needed for the lower bound.

Lemma 3.1 (Angular concentration of diameter neighbours)**.** *Let $x \in S$, and let $y, z \in S \setminus \{x\}$ satisfy*

$$|x - y| = |x - z| = D(S).$$

Then

$$\angle yxz \leq \frac{\pi}{3}.$$

Consequently, all rays from x to its neighbours in $G_D(S)$ are contained in some closed angular arc of width at most $\pi/3$.

Proof. Put $D = D(S)$ and $\theta = \angle yxz \in [0, \pi]$. By the law of cosines,

$$|y - z|^2 = D^2 + D^2 - 2D^2 \cos\theta = 2D^2(1 - \cos\theta).$$

Since D is the diameter of S, one has $|y - z| \leq D$. Hence

$$2(1 - \cos\theta) \leq 1,$$

so $\cos\theta \geq 1/2$ and therefore $\theta \leq \pi/3$.

It remains to justify the final assertion, because pairwise angular separation is a circular rather than a linear condition. Choose one diameter-neighbour direction as angular coordinate 0. Every other neighbour direction has a representative in $[-\pi/3, \pi/3]$. Let a and b be the minimum and maximum of these representatives. If a and b have the same sign, then plainly $b - a \leq \pi/3$. If $a < 0 < b$, then

$$b - a \leq \frac{2\pi}{3} < \pi.$$

Thus $b - a$ is the smaller angular separation between the two extreme directions; by the first part of the proof it is at most $\pi/3$. Hence all neighbour directions lie in the closed arc represented by $[a, b]$, whose width is at most $\pi/3$. $\square$

Fix once and for all an orientation of the plane, so that "clockwise" has an unambiguous meaning. For each non-isolated vertex x of $G_D(S)$, choose a closed arc of width at most $\pi/3$ containing all its diameter-neighbour directions, as supplied by lemma 3.1. Let $c(x)$ be the unique diameter-neighbour whose ray is clockwise-most in that arc. Uniqueness holds because two distinct points at the same positive distance from x cannot lie on the same ray from x.

Lemma 3.2 (Every diameter edge is selected at an endpoint)**.** *If xy is an edge of $G_D(S)$, then*

$$c(x) = y \qquad \textit{or} \qquad c(y) = x.$$

Proof. Suppose for contradiction that neither equality holds. Since y is not clockwise-most among the diameter-neighbours of x, there is a diameter-neighbour y' of x obtained from the ray xy by a strictly positive clockwise turn through some angle

$$0 < \alpha \le \frac{\pi}{3}.$$

Similarly, there is a diameter-neighbour x' of y obtained from the ray yx by a strictly positive clockwise turn through an angle

$$0 < \beta \le \frac{\pi}{3}.$$

Apply a direct similarity of the plane, consisting of a translation, a rotation, and a scaling, so that

$$x = (0,0), \qquad y = (1,0),$$

and the common diameter becomes 1. A clockwise turn from the positive horizontal ray places y' below the x-axis, whereas a clockwise turn from the negative horizontal ray places x' above it. Thus

$$y' = (\cos\alpha, -\sin\alpha), \qquad x' = (1 - \cos\beta, \sin\beta).$$

It follows that

$$\begin{aligned} |y' - x'|^2 - 1 &= (\cos\alpha + \cos\beta - 1)^2 + (\sin\alpha + \sin\beta)^2 - 1 \\ &= 4\cos\frac{\alpha-\beta}{2}\left(\cos\frac{\alpha-\beta}{2} - \cos\frac{\alpha+\beta}{2}\right). \end{aligned}$$

Now

$$\left|\frac{\alpha-\beta}{2}\right| < \frac{\alpha+\beta}{2} \le \frac{\pi}{3}.$$

The cosine function is even and strictly decreasing on $[0,\pi]$. Therefore

$$\cos\frac{\alpha-\beta}{2} > 0 \quad \text{and} \quad \cos\frac{\alpha-\beta}{2} > \cos\frac{\alpha+\beta}{2}.$$

Hence $|y' - x'|^2 - 1 > 0$, so $|y' - x'| > 1$. This contradicts the fact that the diameter is 1. □

Theorem 3.3 (Planar diameter bound). *Every finite set $S \subset \mathbb{R}^2$ with at least two points determines at most $|S|$ diameter pairs:*

$$|E(G_D(S))| \le |S|.$$

Proof. By lemma 3.2, every diameter edge xy has at least one endpoint at which it is the selected edge: either $c(x) = y$ or $c(y) = x$. Assign xy to one such endpoint, choosing arbitrarily if both endpoints select it. A vertex x can receive at most one edge, namely the single edge $xc(x)$. Thus the assignment is an injection from the edge set into the vertex set, and

$$|E(G_D(S))| \le |S|.$$

□

4 The lower bound

Theorem 4.1 (Lower bound). *Let $P \subset \mathbb{R}^2$ be an admissible configuration of $n \ge 2$ points. Then some point $p \in P$ determines at least $\lceil n/3 \rceil$ distinct distances. Equivalently,*

$$f(n) \ge \left\lceil \frac{n}{3} \right\rceil.$$

Proof. By proposition 2.1, every point $p \in P$ satisfies

$$d_P(p) \geq \left\lceil \frac{n-1}{3} \right\rceil.$$

If $n = 3m$ or $n = 3m + 2$, then

$$\left\lceil \frac{n-1}{3} \right\rceil = \left\lceil \frac{n}{3} \right\rceil,$$

so there is nothing more to prove.

It remains to consider $n = 3m + 1$. Suppose, contrary to the desired conclusion, that

$$d_P(p) \leq m \qquad \text{for every } p \in P.$$

The capacity bound gives the reverse inequality $d_P(p) \geq m$, hence

$$d_P(p) = m \qquad \text{for every } p \in P.$$

For a fixed p, the $3m$ points of $P \setminus \{p\}$ are partitioned into exactly m distance fibres, each of cardinality at most 3. Equality of the total cardinalities forces every one of those m fibres to have cardinality exactly 3.

Let

$$D := \max\{|x - y| : x, y \in P\}$$

and let $S \subseteq P$ be the set of all endpoints of pairs at distance D. The set S is nonempty and has at least two points. If $p \in S$, then the distance D is realized from p. Since every distance fibre at p has cardinality exactly 3, the point p has exactly three neighbours at distance D. Every such neighbour also belongs to S. Therefore the diameter graph on S is 3-regular.

The diameter of S is still D: all distances in S are at most D, and S contains the two endpoints of at least one pair at distance D. Hence theorem 3.3, applied to S, gives

$$|E(G_D(S))| \leq |S|.$$

On the other hand, the handshaking identity and 3-regularity give

$$2|E(G_D(S))| = \sum_{p \in S} \deg(p) = 3|S|,$$

so

$$|E(G_D(S))| = \frac{3}{2}|S| > |S|,$$

a contradiction. Thus some $p \in P$ satisfies $d_P(p) \geq m + 1 = \lceil n/3 \rceil$. ☐

Remark 4.2. The diameter argument is needed only when $n \equiv 1 \pmod 3$. It rules out the hypothetical situation in which every distance circle about every point is filled to its maximum permitted capacity of three points.

5 An explicit two-axis construction

We now construct, for every n, an admissible configuration in which no point determines more than approximately $3n/4$ distances.

For an integer $r \geq 0$, define

$$E_r := \begin{cases} \varnothing, & r = 0, \\ \{\pm 2, \pm 2^2, \dots, \pm 2^{r/2}\}, & r \geq 2 \text{ even}, \\ \{1\} \cup \{\pm 2, \pm 2^2, \dots, \pm 2^{(r-1)/2}\}, & r \text{ odd}. \end{cases}$$

Then $|E_r| = r$, every element is nonzero, and E_r is a disjoint union of $\lfloor r/2 \rfloor$ opposite pairs together with one unpaired element when r is odd. In particular, the number of orbits of E_r under $t \mapsto -t$ is

$$\left\lceil \frac{r}{2} \right\rceil.$$

Moreover, the product of any two elements of E_r has the form $\pm 2^k$ for some integer $k \geq 0$.

Given $n \geq 2$, put

$$a := \left\lfloor \frac{n}{2} \right\rfloor, \qquad b := \left\lceil \frac{n}{2} \right\rceil,$$

and define

$$P_n := \big(E_a \times \{0\}\big) \cup \big(\{0\} \times \sqrt{3}\, E_b\big).$$

Because neither E_a nor E_b contains 0, the two displayed parts are disjoint, and therefore $|P_n| = a + b = n$.

The next elementary criterion is the reason for the factor $\sqrt{3}$.

Lemma 5.1 (Four points on two perpendicular axes)**.** *Let u_1, u_2, v_1, v_2 be nonzero real numbers with $u_1 \neq u_2$ and $v_1 \neq v_2$. If the four points*

$$(u_1, 0), \quad (u_2, 0), \quad (0, v_1), \quad (0, v_2)$$

are concyclic, then

$$u_1 u_2 = v_1 v_2.$$

In fact, the equality is also sufficient.

Proof. Every Euclidean circle has an equation

$$x^2 + y^2 + \alpha x + \beta y + \gamma = 0$$

for suitable real coefficients α, β, γ. Restricting this equation to the x-axis shows that u_1, u_2 are the two roots of

$$t^2 + \alpha t + \gamma = 0.$$

Vieta's formula gives $u_1 u_2 = \gamma$. Restricting to the y-axis similarly gives $v_1 v_2 = \gamma$. Hence $u_1 u_2 = v_1 v_2$.

Conversely, if $u_1 u_2 = v_1 v_2 = \gamma$, then

$$x^2 + y^2 - (u_1 + u_2)x - (v_1 + v_2)y + \gamma = 0$$

passes through all four points. Because it passes through two distinct real points, it represents a genuine circle rather than an empty or point circle. Thus the equality is sufficient as well. □

Proposition 5.2 (Admissibility of the construction)**.** *No four points of P_n are concyclic.*

Proof. A circle intersects a fixed line in at most two points, because restricting its equation to that line gives a nonzero quadratic polynomial. Hence a circle containing four points of P_n would have to contain exactly two points on the x-axis and exactly two points on the y-axis.

Write their nonzero coordinates as $u_1, u_2 \in E_a$ and

$$v_1 = \sqrt{3}\, w_1, \qquad v_2 = \sqrt{3}\, w_2, \qquad w_1, w_2 \in E_b.$$

By lemma 5.1, concyclicity would imply

$$u_1 u_2 = v_1 v_2 = 3 w_1 w_2.$$

The left-hand side has the form $\pm 2^r$, while the right-hand side has the form $\pm 3 \cdot 2^s$, for some integers $r, s \geq 0$. Such numbers cannot be equal. If their signs differ, equality is impossible. If their signs agree, their absolute values are unequal because 2^r is not divisible by 3, whereas $3 \cdot 2^s$ is divisible by 3. This contradiction proves the claim. □

Proposition 5.3 (Distance count in the construction)**.** *For $p \in E_a \times \{0\}$,*

$$d_{P_n}(p) \le a - 1 + \left\lceil \frac{b}{2} \right\rceil .$$

For $q \in \{0\} \times \sqrt{3}E_b$,

$$d_{P_n}(q) \le b - 1 + \left\lceil \frac{a}{2} \right\rceil .$$

Proof. Let $p = (u, 0)$ with $u \in E_a$. The other $a - 1$ points on the x-axis contribute at most $a - 1$ distinct distances. For a point $(0, v)$ on the y-axis,

$$|(u, 0) - (0, v)| = \sqrt{u^2 + v^2}.$$

This value is unchanged when v is replaced by $-v$. The set $\sqrt{3}E_b$ has exactly $\lceil b/2 \rceil$ orbits under $v \mapsto -v$, so all b points on the y-axis contribute at most $\lceil b/2 \rceil$ distinct distances from p. Taking the union of the same-axis and cross-axis distance sets can only decrease the sum of these two upper bounds. Therefore

$$d_{P_n}(p) \le a - 1 + \left\lceil \frac{b}{2} \right\rceil .$$

The second inequality is identical after interchanging the two axes. □

Theorem 5.4 (Upper bound)**.** *For every $n \ge 2$,*

$$f(n) \le \left\lfloor \frac{3n}{4} \right\rfloor - \mathbf{1}_{\{4|n\}}.$$

In particular,

$$f(n) \le \left\lfloor \frac{3n}{4} \right\rfloor ,$$

and for every $m \ge 1$,

$$f(4m) \le 3m - 1.$$

Proof. The configuration P_n is admissible by proposition 5.2. By proposition 5.3, its maximum number of distances is at most

$$\max\left\{ a - 1 + \left\lceil \frac{b}{2} \right\rceil , b - 1 + \left\lceil \frac{a}{2} \right\rceil \right\}, \qquad a = \left\lfloor \frac{n}{2} \right\rfloor , \quad b = \left\lceil \frac{n}{2} \right\rceil .$$

The four residue classes of n give the following values.

n	a	b	$a - 1 + \lceil b/2 \rceil$	$b - 1 + \lceil a/2 \rceil$
$4m$	$2m$	$2m$	$3m - 1$	$3m - 1$
$4m + 1$	$2m$	$2m + 1$	$3m$	$3m$
$4m + 2$	$2m + 1$	$2m + 1$	$3m + 1$	$3m + 1$
$4m + 3$	$2m + 1$	$2m + 2$	$3m + 1$	$3m + 2$

Thus the displayed maximum equals

$$\begin{cases} 3m - 1, & n = 4m, \\ 3m, & n = 4m + 1, \\ 3m + 1, & n = 4m + 2, \\ 3m + 2, & n = 4m + 3, \end{cases}$$

which is precisely $\lfloor 3n/4 \rfloor - \mathbf{1}_{\{4|n\}}$. Since $f(n)$ is the minimum over all admissible configurations, the same quantity is an upper bound for $f(n)$. □

Combining theorems 4.1 and 5.4 proves theorem 1.2.

Remark 5.5. The construction deliberately uses many collinear points. This is permitted: the hypothesis excludes four concyclic points but imposes no restriction on collinear subsets.

6 The remaining linear-gap question

The capacity bound treats every occupied circle centred at p as though it could always be filled with three points. The following exact identity measures the failure of that idealized packing.

Definition 6.1. For $p \in P$ and $j \in \{1, 2, 3\}$, let $N_j(p)$ be the number of realized distances r for which

$$|F_p(r)| = j.$$

Define the distance-fibre deficiency at p by

$$\Delta_P(p) := 2N_1(p) + N_2(p).$$

Proposition 6.2 (Exact deficiency identity)**.** *For every $p \in P$,*

$$\boxed{3d_P(p) - (n-1) = \Delta_P(p).}$$

Equivalently,

$$d_P(p) = \frac{n - 1 + \Delta_P(p)}{3}.$$

Proof. The distance fibres partition $P \setminus \{p\}$, and every fibre has size 1, 2, or 3. Therefore

$$n - 1 = N_1(p) + 2N_2(p) + 3N_3(p),$$

whereas

$$d_P(p) = N_1(p) + N_2(p) + N_3(p).$$

Subtracting the first identity from three times the second gives

$$3d_P(p) - (n-1) = 2N_1(p) + N_2(p) = \Delta_P(p).$$

□

Corollary 6.3 (Exact reformulation of a linear improvement)**.** *Fix $c > 0$. For a given admissible n-point configuration P, the assertion that some $p \in P$ satisfies*

$$d_P(p) > \left(\frac{1}{3} + c\right) n$$

is equivalent to

$$\max_{p \in P} \Delta_P(p) > 3cn + 1.$$

Consequently, a universal estimate

$$f(n) > \left(\frac{1}{3} + c\right) n$$

for all sufficiently large n is equivalent to proving that every sufficiently large admissible configuration has a point whose distance-fibre deficiency is larger than $3cn + 1$.

Proof. By proposition 6.2,

$$\begin{aligned} d_P(p) > \left(\frac{1}{3} + c\right) n &\iff n - 1 + \Delta_P(p) > n + 3cn \\ &\iff \Delta_P(p) > 3cn + 1. \end{aligned}$$

Taking the maximum over $p \in P$ proves the first assertion. Taking the minimum over all admissible configurations proves the second. □

Thus the second question in the problem is not decided by the present bounds. A positive constant improvement over $1/3$ requires a genuinely global argument showing that, for at least one common centre, a positive proportion of the available three-point capacities remain unused. The diameter-graph argument supplies only the constant-sized improvement needed in the residue class $n \equiv 1 \pmod 3$.

7 Small cases and final summary

The first few exact values follow immediately:

$$f(1) = 0, \qquad f(2) = 1, \qquad f(3) = 1, \qquad f(4) = 2.$$

Indeed, the first two are immediate; an equilateral triangle gives $f(3) = 1$; and theorem 1.2 gives $2 \leq f(4) \leq 2$.

The unconditional conclusions established in this note are therefore

$$\boxed{\left\lceil \frac{n}{3} \right\rceil \leq f(n) \leq \left\lfloor \frac{3n}{4} \right\rfloor - \mathbf{1}_{\{4|n\}}.}$$

The upper construction disproves the near-linear conjecture $f(n) > (1 - o(1))n$. Whether the lower bound admits a uniform improvement to $(1/3 + c)n$ for some absolute $c > 0$ is not resolved by these arguments; corollary 6.3 records the exact structural statement that such an improvement would require.